Adult Income Prediction — Day 3 Write-Up

Engineered Features

I added eight new features. Each one uses only values from the same row (no target encoding), so they fit cleanly inside the CV pipeline without leakage.

Feature	Type	Rule	Why it might help	Univariate signal (train)
age_bucket	categorical	bins at 25/35/45/55/65	Mid-career ages earn more	MI=0.063; >50K rate 2% (<=25) to 39% (46–55)
hours_bin	categorical	part-time / full / overtime / extreme	Work intensity is non-linear	MI=0.038; rate 7% (part-time) to 41% (extreme)
has_capital_gain	binary	capital-gain > 0	Most people have zero gain	MI=0.032; rate 0.62 vs 0.21
log_capital_gain	numeric	log1p(capital-gain)	Compresses heavy skew	MI=0.084 (highest)
is_higher_edu	binary	education-num >= 13	Same threshold as Day 1 rule	MI=0.052; rate 0.48 vs 0.16
edu_x_hours	numeric	education-num × hours	Interaction of skill and effort	MI=0.081; quintile spread ~0.45
has_capital_loss	binary	capital-loss > 0	Rare investment activity flag	MI=0.012; rate 0.50 vs 0.23
is_full_time	binary	hours >= 40	Simple full-time vs not flag	MI=0.026; rate 0.28 vs 0.09

I also considered a few options and skipped them: target encoding (easy to leak labels without nested CV), mean imputation (pulled around by outliers), dropping rows with missing values (loses data), and LabelEncoder (pretends categories are ordered).

Cross-Validated Model Comparison

I used stratified 5-fold CV on the training set only, with the same Day 1 hold-out split left aside. All three models shared the same feature engineering and preprocessing steps.

Model	F1 (mean ± std)	ROC AUC (mean ± std)	Accuracy mean
HistGradientBoosting	0.7084 ± 0.0094	0.9273 ± 0.0026	~0.87
Logistic Regression	0.6755 ± 0.0108	0.9136 ± 0.0033	~0.86
Random Forest	0.6616 ± 0.0085	0.9009 ± 0.0047	~0.85

Day 2 logistic regression had hold-out F1 = 0.6630. With the new features, CV F1 is about 0.676 for logistic regression and 0.708 for HistGradientBoosting — a clear improvement before any tuning. In the notebook boxplots, HGB stays ahead on both F1 and ROC AUC across folds.

Statistical Comparison

The top two models by F1 were HistGradientBoosting and Logistic Regression.

- Mean F1 difference: 0.0329
- Paired t-test: $p < 0.001$
- Wilcoxon signed-rank: $p \approx 0.06$ (only 5 folds, so this test is weak; HGB won every fold)

A roughly 3-point F1 gap is large enough that I will tune HistGradientBoosting first in Day 4. Logistic regression is still useful as a faster, more interpretable backup — especially for trying `class_weight='balanced'` later.

Engineered features that mattered

- **Random Forest:** `edu_x_hours`, `log_capital_gain`, and `is_higher_edu` ranked highest among the new columns.
- **Logistic Regression:** `log_capital_gain` and `has_capital_gain` dominate (they overlap), then age buckets (young/old negative, mid-career positive) and part-time hours (negative).

That matches what we saw in Day 1 and Day 2: capital activity, education, and hours/life stage matter together. Education alone was never enough.

Feature Selection Decision

Setup	F1 mean	Notes
All features	0.7084	Strong result
SelectKBest k=100	0.7088	Almost the same; MI scoring takes longer
SelectKBest k=50	0.6990	Clear drop

For Day 4 I will keep all eight engineered features and the full feature set for HistGradientBoosting. Cutting to 50 features hurt F1, and k=100 did not really beat the full set.

Day 4 plan: tune HGB (learning rate, depth/leaf settings, iterations). I may also retry logistic regression with `class_weight='balanced'` and test dropping either `education` or `education-num`. The hold-out test set stays unused until after tuning.